Student Guide: Vibe Check: Build Your Money Brain

FinTech Club | Event 1 of 5 | Friday, October 23 | 3:00 to 5:00 PM | TEC 202

Welcome

Welcome to the FinTech Innovation Series! Over five events, your team will build an AI-powered personal finance dashboard: one app that shows your accounts, spending, and alerts in one place, and lets you ask an AI questions about your money. Event 1 is the kickoff. You will meet your team, pick a role, plan the app, and get a working starter project running.

You do not need coding or finance experience. Nobody here is an expert, and we are learning together. All data in this project is fake, so no real money or accounts are ever involved.

Installation

Option A: Install on your own computer (recommended)

Windows

1. Download Python 3.11 or 3.12 from python.org/downloads. On the first installer screen, check the box Add python.exe to PATH, then click Install Now.
2. Open Command Prompt and type `python --version`. You should see a version number.

Mac

1. Download Python 3.11 or 3.12 from python.org/downloads and run the installer.
2. Open Terminal and type `python3 --version`. You should see a version number.

Both

3. Install VS Code, a free code editor, from code.visualstudio.com.
4. Download the starter project: on the starter repo page click Code, then Download ZIP. Unzip it and open the folder in VS Code (File, then Open Folder).
5. Open a terminal in VS Code (Terminal, then New Terminal) and run:
    - Windows: `python -m pip install -r requirements.txt` and then `python -m streamlit run app.py`
    - Mac: `python3 -m pip install -r requirements.txt` and then `python3 -m streamlit run app.py`
6. Your browser should open the dashboard, usually at http://localhost:8501.

Option B: Nothing to install (optional)

GitHub Codespaces is a coding environment that runs in your web browser, so you do not install Python or anything else.

1. Go to the club's starter project: [STARTER REPO LINK]
2. Click the green Code button, open the Codespaces tab, and click Create codespace on main. (If GitHub asks to fork the project, say yes.)
3. Wait one to two minutes while it sets up.
4. In the Terminal panel at the bottom, type `streamlit run app.py` and press Enter.
5. Click Open in Browser when the pop-up appears. You should see a dashboard with fake accounts.
6. When you are done, press Ctrl+C in the terminal to stop the app, then stop your Codespace from github.com/codespaces so it does not use up your free hours.

Codespaces has a free monthly allowance for personal accounts that should cover this series. Check GitHub's current limits, and look into the GitHub Student Developer Pack, which may give students more.

Troubleshooting

Problem	Fix
python is not recognized (Windows)	Reinstall Python and check the PATH box, or try `py --version`
streamlit is not recognized	Use `python -m streamlit run app.py` (or `python3` on Mac)
pip is not recognized	Use `python -m pip install -r requirements.txt`
Mac error: externally-managed-environment	Use Codespaces (Part A) instead
Browser did not open	Go to http://localhost:8501. In Codespaces, open the Ports tab and click the globe icon
Nothing works	No stress. Pair up with a teammate, and ask a mentor at the event

Roles

Each team has 3 to 4 people. If your team is small, one person can cover two roles. Roles can change in later events.

- **Developer:** owns the code and the team repo. Event 1: set up the repo and run the app. Later: connect to the bank, wire in the AI, fix security issues.
- **Designer:** owns the look and flow of the app. Event 1: sketch the main screen. Later: layout, charts, polish, demo slides.

- **AI Prompter:** owns the assistant. Event 1: list the questions users would ask about their money. Later: write and test the prompts.
- **Researcher:** owns the why and the what could go wrong. Event 1: look at existing money apps and build the feature list. Later: the security checklist and the demo script.

Core tasks

Task 1: Meet your team (10 minutes)

Introduce yourselves. Then pick a team name, because you will need it for your repo.

Done when: you have a team name.

Task 2: Pick roles (5 minutes)

Each person picks a main role from the Roles section. In a team of three or less, combine roles (for example Designer plus Researcher).

Done when: every role has an owner written down.

Task 3: Write your feature list (about 25 minutes)

Decide what your dashboard should do. Write 3 must-haves and 2 nice-to-haves. Phrase each one as: A user can ___.

Ideas to get you started:

- See all accounts and balances in one place
- See spending by category each month
- Ask the AI, Where did my money go this month?
- Get an alert for an unusual charge
- Track subscriptions
- Set a savings goal

Researcher: look at two existing money apps and note one thing each does well.

AI Prompter: write 5 questions a real person would ask about their money.

Done when: 5 features and 5 questions are on your team sheet.

Task 4: Sketch the main screen (about 30 minutes)

Draw the main screen of your dashboard on paper, a whiteboard, Canva or Figma. Draw boxes and label them: account summary, transactions list, Ask AI box, and alerts. Do not worry about making it pretty. Placement and clarity matter more.

Done when: you have a sketch of the main screen, and a photo of it saved to your team sheet.

Task 5: Set up your repo and run the app (about 30 minutes)

Teammates are welcome to follow along on their own screens.

1. Open the starter project: [STARTER REPO LINK]. Click the green Use this template button, then Create a new repository.
2. Name it with your team name plus -dashboard. Choose Private or Public, then create it.
3. Add teammates: Settings, then Collaborators, then Add people, and enter their GitHub usernames.
4. In your new repo, click Code, open the Codespaces tab, and click Create codespace on main.
5. In the terminal, run `pip install -r requirements.txt` (it may already be done for you), then `streamlit run app.py`.
6. Click Open in Browser. You should see the dashboard with fake accounts and transactions.
7. Make your first change: open app.py, find the line marked TEAM TODO, and change the title to your team name. Save the file (Ctrl+S, or Cmd+S on Mac). The page updates in your browser.
8. Save your work to GitHub: click the Source Control icon on the left, type a message such as Add team name, click Commit, then click Sync Changes.

Done when: your dashboard shows your team name, and the change appears in your repo on GitHub.

Task 6: Share out (about 20 minutes)

Each team gets 60 seconds to present: your team name, who has which role, your top 3 features, your sketch, and a quick look at the running app.

Done when: you have presented.

Bonus challenges

Bonus challenges are optional and only for teams that have finished Tasks 1 to 5. Nothing in Events 2 to 5 depends on them.

Silver challenges

- **Name the app:** Choose an app name, a one-line tagline, and 2 or 3 brand colors.
- **Customize the app:** In app.py, change the app title and caption, and add one more fake account to the accounts table.
- **Competitor check:** Find two real money apps. For each, write one thing they do well and one gap your app could fill.

Gold challenges

- **Add a chart:** Add a bar chart of fake spending by category to the dashboard. Hint: the starter project already has the transactions table, and Streamlit has a built-in st.bar_chart.
- **Write the assistant's rules:** Write 5 rules for your future AI assistant. For example, what it should always do, and what it should refuse, such as moving money or guaranteeing investment returns.
- **What could go wrong:** If this app handled real bank data, list 3 things that could go wrong and how you would prevent each one. This sets you up for Event 4.

Ground rules

- **Fake data only.** Never type real bank logins, card numbers, or personal ID numbers into the project or into any AI tool.
- **No secrets in code.** Never paste passwords or API keys into your code or onto GitHub.
- **Help each other.** There are no silly questions. If you know it, explain it. If you do not, ask.
- **Educational only.** This project is for learning and is not financial advice.

After the event: before Event 2 (Friday, October 30)

Event 2 is Plug It In: Connecting to the Bank (Safely). Your team will connect to a sandbox bank with fake accounts.

- Keep your team sheet and sketch somewhere your whole team can find them.
- Developer: make sure your desktop app or Codespace still opens and runs.
- Optional head start: create a free developer account at plaid.com. We will walk through it together at Event 2, so you can skip this.
- Everyone: think of one real money question you wish an app could answer.